

- 4 (a) Using a light microscope at a magnification of $\times 400$, it is possible to identify different types of blood cell in prepared slides of mammalian blood.

Fig. 4.1 is a key to identify different types of blood cell in prepared slides of mammalian blood.

In Fig. 4.1, letters **C**, **D**, **E** and **F** represent four different types of blood cell.

key	
1a nucleus present	go to 2
1b nucleus absent	C
2a large rounded (spherical) nucleus	D
2b nucleus not rounded	go to 3
3a nucleus is kidney shaped	E
3b nucleus is lobed	F

Fig. 4.1

- (i) Identify the cell types **C**, **D**, **E** and **F** in Fig. 4.1.

C

D

E

F

[3]

- (ii) Explain why blood taken from a person with an infectious disease may have a different number of white blood cells compared with blood taken from a healthy person.

.....

.....

.....

.....

..... [2]

As part of a research project, a team of doctors and scientists carried out tests to investigate the effect of altitude on the ability of blood to carry oxygen. They assessed the changes in the oxygen-carrying ability of their own blood as they walked from 1530m above sea level to 5700m above sea level, over a period of 14 days. 5700m above sea level is an extremely high altitude.

Table 4.1 is a summary of some of their results.

Table 4.1

	day 1 at 1530 m	day 14 at 5700 m
percentage saturation of haemoglobin with oxygen in blood leaving the lungs	95	82
volume of oxygen delivered to tissues by 100cm ³ of blood/cm ³	19.3	19.2

- (b) Compare the results for day 1 with the results for day 14 in Table 4.1 **and** explain the results shown for **day 14**.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (c) The results of the research project have led to an improved understanding of diseases such as sickle cell anaemia.

Explain why sickle cell anaemia is an example of a disease.

.....

.....

.....

.....

..... [2]

[Total: 11]